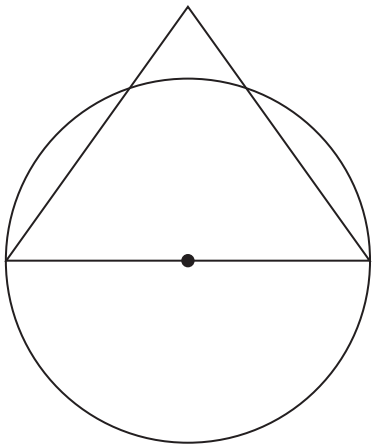
13 The diagram shows a circle and an equilateral triangle.



One side of the equilateral triangle is a diameter of the circle.
The circle has a circumference of 44 cm.

Work out the area of the triangle.
Give your answer correct to 3 significant figures.

Circumference = πd

$44 = \pi d$, so $d = 44 \div \pi$

$d = 14.00563499...$ cm

1 mark
Find the side length from the circumference

Equilateral triangle: every side is d, every angle is 60°.

$$\text{Area} = \frac{1}{2} \times d \times d \times \sin 60°$$

1 mark
Use ½ab sin C

$= \frac{1}{2} \times (44 \div \pi)^2 \times \sin 60°$

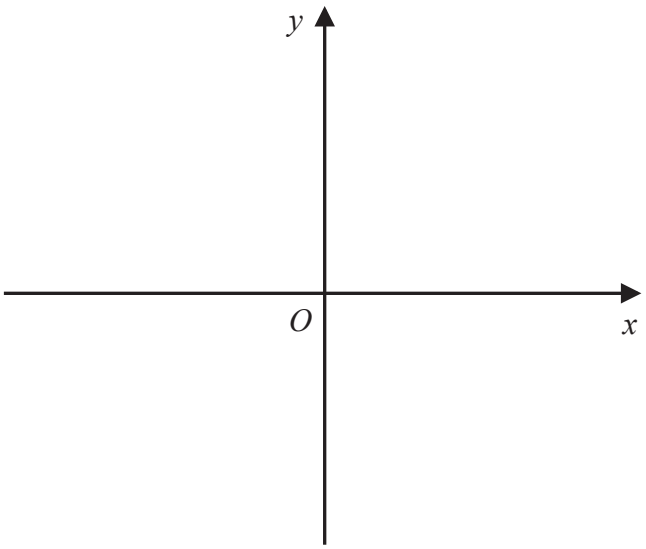
$= 84.9388239...$ cm^2

1 mark
Round only the final answer to 3 significant figures

84.9 cm^2

(Total for Question 13 is 3 marks)

14 On the grid, sketch the curve with equation $y = 2^x$
Give the coordinates of any points of intersection with the axes.



(Total for Question 14 is 2 marks)

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